

HERMES: Healthcare Ethics Robustness in Medical Image Systems

Enhancing Robustness and Ethical Integrity of Image Classifiers Against Adversarial Attacks

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Abstract

Deep Neural Networks (DNNs) have shown exceptional promise in medical image analysis, yet their vulnerability to adversarial attacks poses a critical threat to patient safety and clinical trust. This paper, HERMES (Healthcare Ethics Robustness in Medical Image Systems), presents a systematic investigation into the robustness of medical image classifiers under a realistic gray-box threat model. We fine-tune a ResNet-18 model on the PatchCamelyon (PCam) dataset for metastatic tissue detection and evaluate its resilience against two potent evasion attacks: the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD). To counter these threats, we implement and benchmark a suite of input preprocessing defenses, including Gaussian smoothing, JPEG compression, spatial smoothing, and total variance minimization. Our analysis rigorously assesses each defense's effectiveness in preserving accuracy on both benign and adversarial images, while also considering computational overhead. We further explore the impact of standard data augmentation on the model's intrinsic robustness. By providing a comparative analysis of attack efficacy and defense performance, this work offers crucial insights for developing more secure, reliable, and ethically sound AI systems for clinical deployment, ultimately aiming to enhance trust in automated diagnostic tools. The code for reproducing the results is available at <https://github.com/MatteoFasulo/HERMES>.

Keywords

Adversarial Robustness, Medical Image Analysis, Adversarial Defenses, Gray-box Attacks, AI Ethics

1. Introduction

Deep Neural Networks (DNNs) have emerged as a transformative technology in medical image analysis, demonstrating remarkable capabilities for automated diagnosis and clinical decision support [1]. These models can process vast quantities of medical imaging data—spanning modalities from radiology and histopathology to ophthalmology—with a precision that often rivals or surpasses that of human experts. This breakthrough holds immense potential to enhance healthcare delivery by enabling earlier disease detection, improving diagnostic consistency, and expanding access to expert-level analysis in resource-limited environments.

Despite their success, DNNs exhibit a critical vulnerability to adversarial attacks: subtle, often imperceptible perturbations to input data designed to elicit misclassifications. In the high-stakes domain of medical diagnostics, where an incorrect prediction can have life-threatening consequences, this vulnerability is particularly alarming. These malicious alterations, known as *evasion attacks*, are typically undetectable by human specialists, undermining the reliability of AI-driven systems. The potential for such failures erodes trust among clinicians and patients, posing a significant barrier to the widespread adoption of these powerful technologies in clinical practice.

To mitigate these risks, a variety of defense strategies have been proposed. Our work operates under the **gray-box threat model**, a realistic scenario where an adversary possesses partial knowledge of the target system, such as its architecture (e.g., ResNet) and parameters, but remains unaware of specific defense mechanisms in place. This model represents a credible threat, as attackers may gain access to model details through public APIs, leaked information, or open-source implementations.

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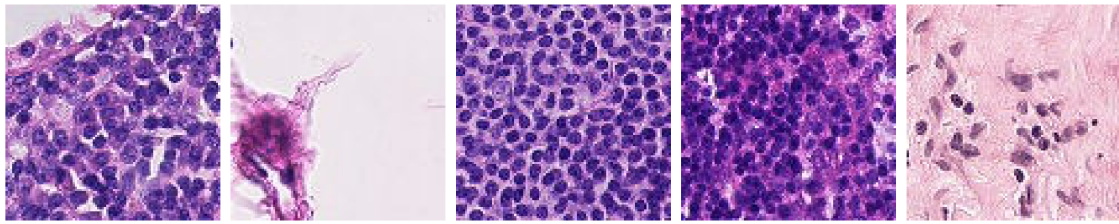


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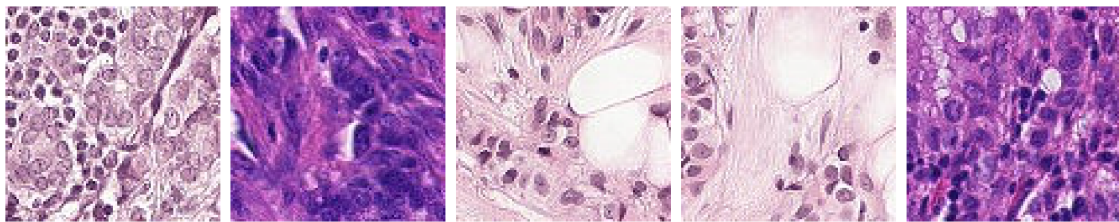
In this report, we conduct a systematic investigation into the robustness of a medical image classifier against gray-box adversarial attacks. We fine-tune a ResNet-18 architecture [2] on the PatchCamelyon (PCam) dataset [3] for metastatic tissue detection. We then evaluate its vulnerability to two prominent evasion attacks: the Fast Gradient Sign Method (FGSM) [4] and Projected Gradient Descent (PGD) [5]. To counter these threats, we implement and benchmark a suite of preprocessing defenses, including Gaussian smoothing, JPEG compression [6], spatial smoothing [7], and total variance minimization [8]. Our evaluation assesses each defense based on its ability to preserve classification accuracy on both benign and adversarial samples, alongside its computational overhead. Finally, we provide an ethical analysis of our findings, discussing the implications for patient safety, system reliability, and trust in medical AI. This work aims to provide foundational insights for developing more secure and clinically viable AI systems.

2. Dataset

Our study utilizes the PatchCamelyon (PCam) dataset [3], a publicly available collection of histopathologic scans of lymph node sections. PCam is derived from the Camelyon16 challenge dataset and is specifically designed for binary classification: identifying whether a given image patch contains metastatic tissue. The dataset consists of 327,680 color image patches of 96×96 pixels each, extracted from whole-slide images. Each patch is labeled as either positive (containing at least one pixel of tumor tissue in the center 32×32 region) or negative (healthy tissue). Representative samples are shown in Figure 1. The balanced nature and large scale of PCam make it an excellent benchmark for training and evaluating deep learning models for medical image classification.



(a) Negative samples (healthy tissue without metastasis).



(b) Positive samples (tissue containing metastatic cells).

Figure 1: Representative samples from the PatchCamelyon dataset [3] showing histopathologic image patches (96×96 pixels). The task is to classify each patch as containing metastatic cancer cells or not.

We chose PatchCamelyon dataset for several key reasons. First, it addresses a genuine clinical challenge in cancer diagnosis where metastatic tissue detection is critical for patient treatment decisions. This ensures our adversarial robustness findings have direct implications for medical AI deployment. Second, the dataset is a good balance between complexity and computational tractability, being more challenging than standard benchmarks like CIFAR-10 while remaining manageable for extensive experimentation with multiple attack and defense methods on a limited GPU hardware.

The binary classification nature provides clear interpretability for adversarial impact assessment, as misclassifications directly translate to clinically meaningful errors with significant consequences.

3. Model Architectures and Training

We employed ResNet-18 [2], as it provides an optimal balance between computational efficiency and classification performance. This architecture is well-suited for training with limited GPU resources while demonstrating excellent performance across various computer vision domains, including medical imaging applications.

To evaluate the impact of data augmentation on adversarial robustness, we conducted two distinct training experiments. The first model was trained without any data augmentation techniques. The second model was trained with standard data augmentation, including random horizontal and vertical flips, rotations, and color jittering. This approach aims to introduce robustness to roto-translations and partial variations in illumination and occlusion. By comparing these two models, we can benchmark whether standard augmentation practices enhance resilience against adversarial attacks, a crucial consideration for the safety of medical AI systems.

4. Attacks

To evaluate the robustness of our trained models, we utilized two widely studied adversarial attack methods: **FGSM** [4] and **PGD** [5]. Both are gradient-based white-box attacks, meaning the attacker requires full access to the model's parameters and gradients to craft the adversarial examples.

4.1. Gray-Box Attack Scenario

Our evaluation is framed within a **gray-box attack scenario**, which simulates a realistic threat where an adversary has partial knowledge of the target system. In this model, the attacker has access to the ResNet-18 network architecture and the trained model parameters but remains unaware of any specific defense mechanisms, such as input preprocessing, that have been implemented. This setup is depicted in Figure 2.

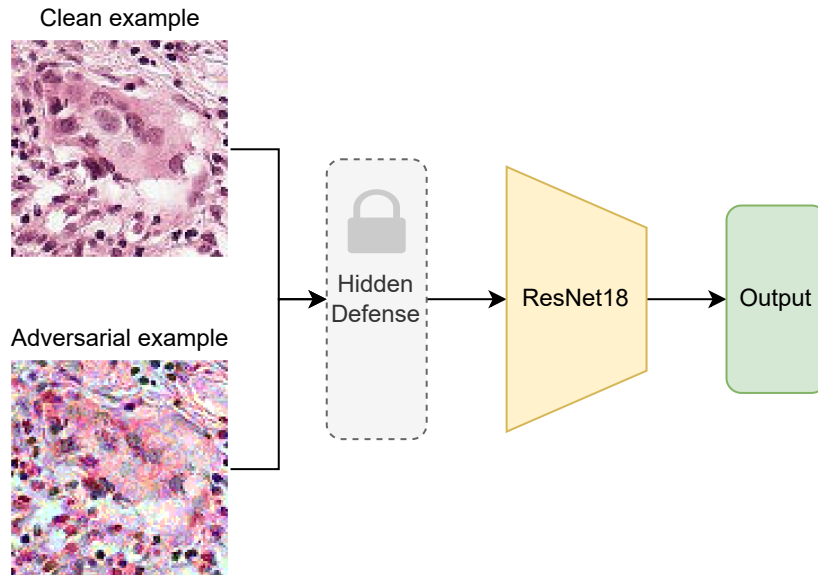


Figure 2: A diagram of the gray-box attack pipeline. The attacker crafts an adversarial example using knowledge of the model but is unaware of the defense layer that preprocesses the input before classification.

The procedure for generating and evaluating adversarial examples in this scenario is as follows:

1. **Input Selection:** An attacker selects a clean input sample from the dataset to serve as the base for the adversarial example.
2. **Perturbation Crafting:** Leveraging knowledge of the target model’s architecture and weights, the attacker computes the gradient of the loss function with respect to the input image. This gradient is used to craft a small perturbation that, when added to the original image, is designed to maximize the model’s loss and induce a misclassification. The resulting adversarial example remains visually indistinguishable from the original.
3. **Submission to Target System:** The crafted adversarial image is submitted to the target system for classification. The attacker has no further insight into the system’s internal operations.
4. **Defense Processing:** Before the image is fed to the classifier, it passes through a concealed defense mechanism. This module, which acts as a preprocessing step, attempts to purify the input by mitigating or removing the adversarial perturbation. The attacker has no knowledge of this defense.

4.2. Fast Gradient Sign Method (FGSM)

The Fast Gradient Sign Method is a single-step attack that perturbs an input in the direction of the sign of the loss function’s gradient. The goal is to efficiently generate an adversarial example by adding a small, structured noise pattern that maximizes the model’s prediction error.

Let x be the original input image, y be the true label, θ be the model parameters, and $J(\theta, x, y)$ be the loss function. The adversarial example x^{adv} is generated as:

$$x^{\text{adv}} = x + \varepsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

where ε is a small scalar that controls the magnitude of the perturbation. A larger ε increases the attack’s potency at the cost of making the perturbation more visually apparent.

4.3. Projected Gradient Descent (PGD)

Projected Gradient Descent is an iterative extension of FGSM and is considered one of the strongest first-order adversarial attacks. PGD applies multiple small, FGSM-like steps, and after each step, it projects the perturbed image back into an ε -ball around the original input, typically defined by the L_∞ norm. An ε -ball in the L_∞ metric is the set of all points x' such that for every pixel coordinate i , $|x_i - x'_i| \leq \varepsilon$.

Let $\mathcal{B}_\varepsilon(x)$ be the ε -ball around x and let $\Pi_{\mathcal{B}_\varepsilon(x)}(\cdot)$ be the projection operator onto this ball. The update rule for each iteration t is given by:

$$x_{t+1} = \Pi_{\mathcal{B}_\varepsilon(x)}(x_t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x_t, y)))$$

where x_0 is the original image x (or a random point within the ε -ball), and α is the step size (typically $\alpha < \varepsilon$). This iterative process allows PGD to find more effective adversarial examples by exploring the loss landscape more thoroughly than single-step methods like FGSM.

5. Defenses

To mitigate the impact of adversarial attacks, we implemented and evaluated several input preprocessing techniques. These defenses aim to remove or reduce adversarial perturbations before the input reaches the classification model. The methods investigated are: **Gaussian Smoothing**, **Spatial Smoothing**, **Total Variation Minimization**, and **JPEG Compression**.

5.1. Gaussian Smoothing

5.1.1. Introduction

Gaussian smoothing is a simple yet effective denoising technique that reduces high-frequency noise in an image. It operates by convolving the input with a Gaussian kernel, which averages pixel values with their neighbors, weighted by a Gaussian function.

Let x be the input image and G_σ be a Gaussian kernel with standard deviation σ . The smoothed image x^{smooth} is computed as:

$$x^{\text{smooth}} = x * G_\sigma$$

where $*$ denotes the convolution operation. Larger values of σ result in stronger smoothing but also increase image blur, potentially removing important features.

5.1.2. Defense against FGSM Adversarial Attack

We evaluate the effectiveness of Gaussian smoothing as a defense by applying it to adversarial images generated by FGSM before classification. Figure 9 illustrates the trade-off between robustness and accuracy on clean data as the smoothing parameter (σ) is varied for models trained with and without data augmentation.

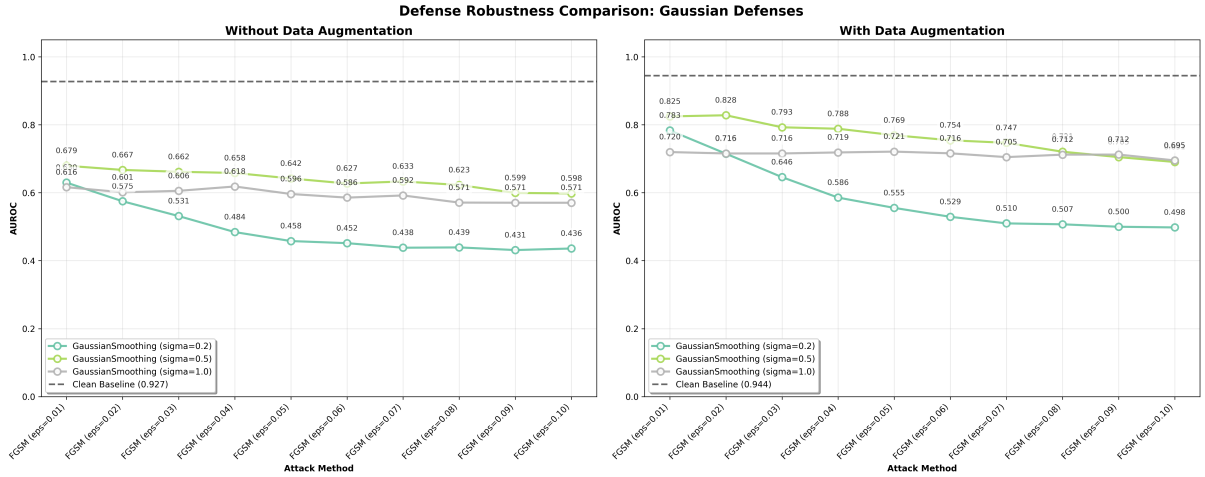


Figure 3: Classification accuracy under FGSM adversarial attacks with Gaussian smoothing defense. Results compare performance with and without data augmentation across different sigma values

Figure 9 illustrates the performance of Gaussian smoothing against FGSM attacks. The model trained with data augmentation (right) demonstrates substantially higher robustness, achieving a consistently better AUROC than the non-augmented model (left) across all attack strengths. The choice of the hyperparameter sigma is crucial: a value of sigma=0.5 (yellow-green line) offers the optimal defense by balancing noise removal and feature preservation. In contrast, weaker smoothing (sigma=0.2, teal line) is insufficient to counter the attack, while stronger smoothing (sigma=1.0, gray line) degrades performance by excessively blurring the image, thereby harming classification accuracy.

5.1.3. Defense against PGD adversarial attack

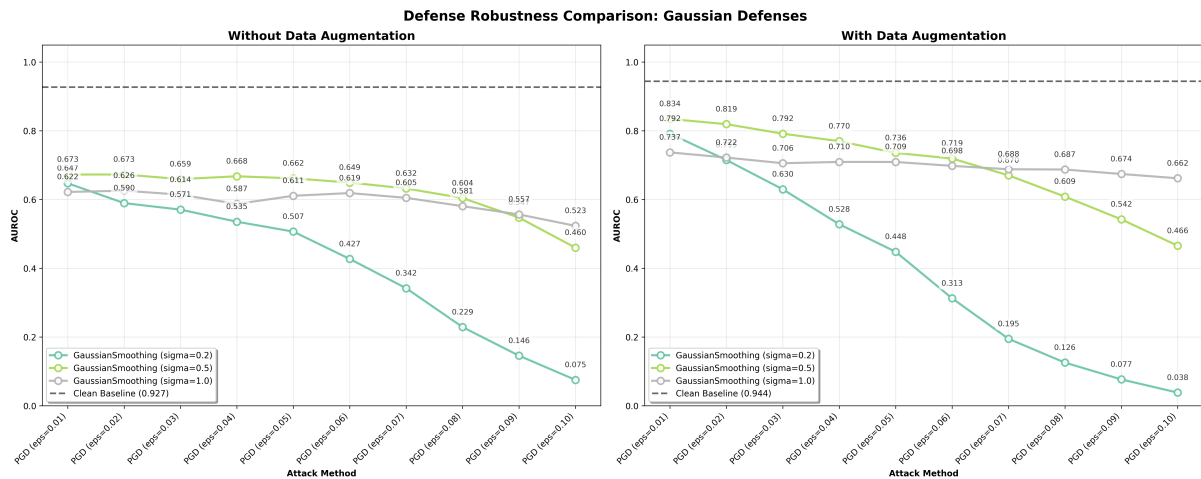


Figure 4: Classification accuracy under PGD adversarial attacks with Gaussian smoothing defense. Results compare performance with and without data augmentation across different sigma values

5.1.4. Impact on the baseline performance

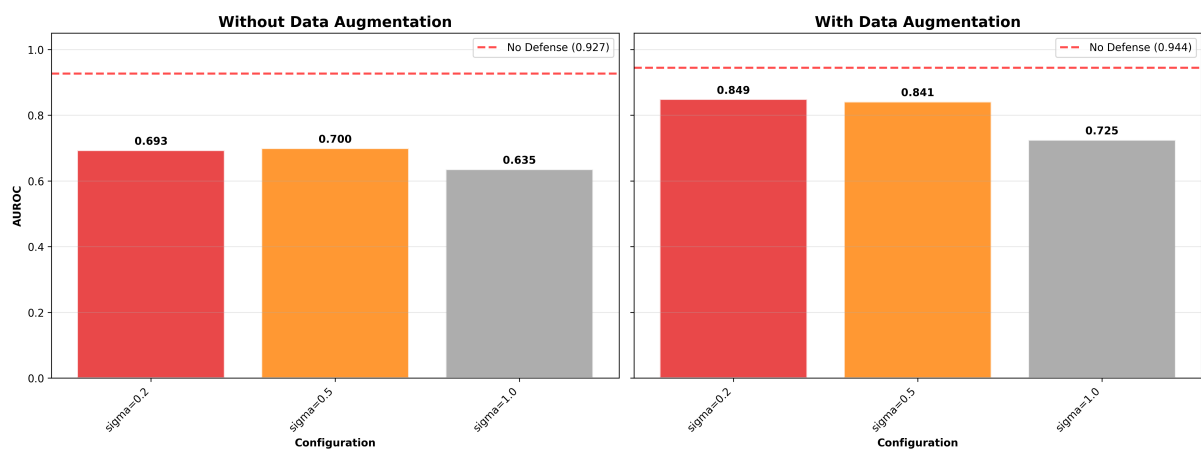


Figure 5: Impact of Gaussian smoothing defense on clean images. The plot shows a baseline when no defense is applied and the AUROC when using different sigma values with and without data augmentation

5.2. Spatial Smoothing

Specifically, median filtering, is another denoising technique that replaces each pixel's value with the median of its neighboring values. This approach is particularly effective at removing salt-and-pepper type noise and other types of outlier pixels.

For every pixel with coordinates (i, j) , define its neighborhood as $\mathcal{N}_{i,j}$. Then, smoothed pixel $x_{i,j}^{smooth}$ is computed as:

$$x_{i,j}^{smooth} = \text{median}(x_{m,n} | (m,n) \in \mathcal{N}_{i,j})$$

5.2.1. Defense against FGSM Adversarial Attack

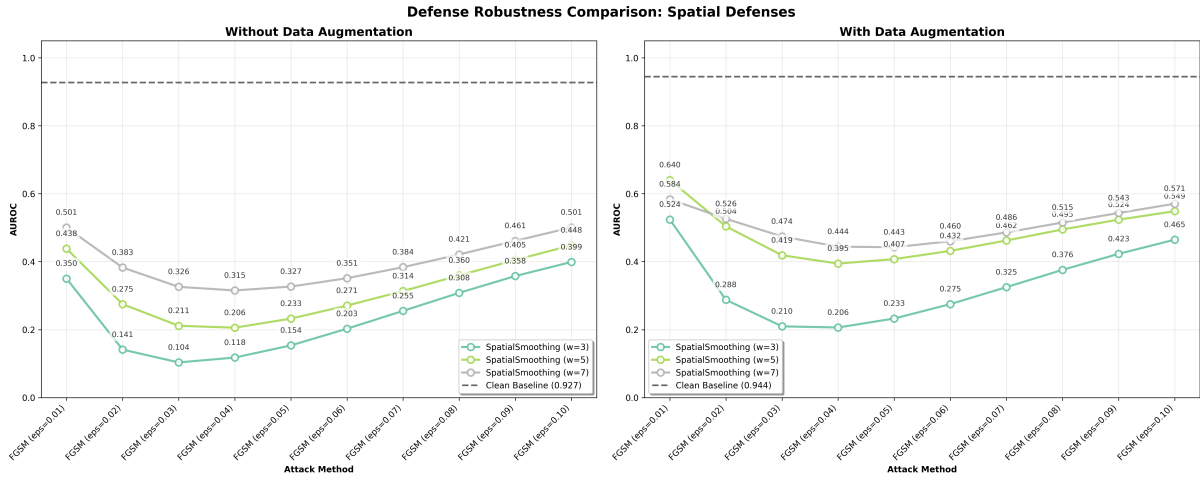


Figure 6: Classification accuracy under FGSM adversarial attacks with spatial smoothing defense. Results compare performance with and without data augmentation across different window sizes

Figure 6 evaluates the performance of spatial smoothing (median filtering) as a defense against FGSM attacks. Overall, this defense proves to be less effective than Gaussian smoothing, with the AUROC dropping significantly even at low attack strengths. The model trained with data augmentation (right) again shows superior baseline robustness compared to the non-augmented model (left). A larger filter window ($w=7$, gray line) generally offers the best protection among the tested configurations. A notable and counter-intuitive trend is the recovery of performance at higher epsilon values ($\text{eps} > 0.04$). We attribute this phenomenon less to the defense's properties and more to the fundamental nature of the FGSM attack itself. FGSM performs a single, linear step in the direction of the gradient's sign. This method was originally conceived to demonstrate the locally linear behavior of neural networks. While this single step effectively increases the loss, it is not guaranteed to be the optimal path toward a misclassification, especially at large magnitudes. As epsilon increases, the FGSM perturbation may "overshoot" the most vulnerable regions of the decision boundary, landing the perturbed image in a less effective adversarial space. In contrast, an iterative method like PGD explores the loss landscape more thoroughly, taking multiple smaller steps to find a point that consistently maximizes the loss. Therefore, the observed accuracy recovery likely highlights a limitation of the single-step FGSM attack at high strengths, rather than an increased efficacy of the spatial filter against coarse perturbations.

5.2.2. Defense against PGD adversarial attack

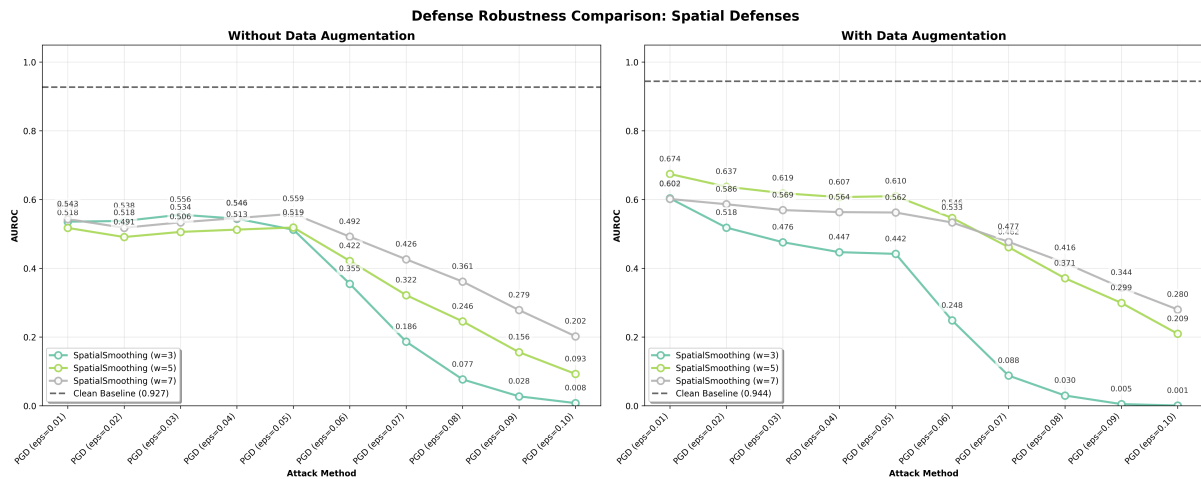


Figure 7: Classification accuracy under PGD adversarial attacks with Spatial smoothing defense. Results compare performance with and without data augmentation across different sigma values

5.2.3. Impact on the baseline performance

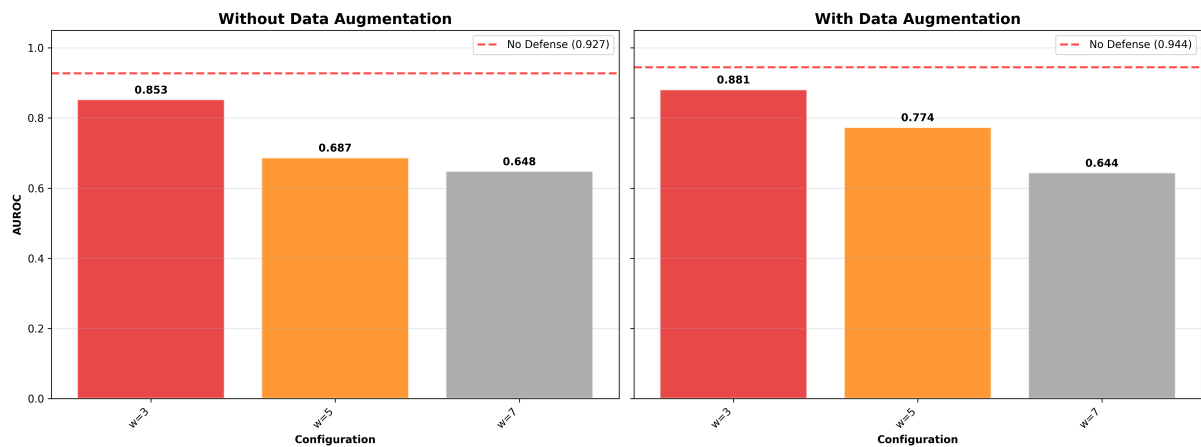


Figure 8: Impact of spatial smoothing defense on clean image classification accuracy. The plot compares performance on unprocessed clean images (red dotted line) versus images preprocessed with spatial smoothing at different window sizes

5.3. Total Variation Minimization

Reduces the total variation in the image, effectively removing small, high-frequency components such as adversarial noise while preserving image edges. Total variation of an image x is defined as:

$$TV(x) = \sum_{i,j} \sqrt{(x_{i+1,j} - x_{i,j})^2 + (x_{i,j+1} - x_{i,j})^2}$$

To defend against attacks, the goal is to find a denoised version of x that is close to the original image but with minimal total variation:

$$x' = \operatorname{argmin}_z ||z - x||_2^2 + \lambda \cdot TV(z)$$

Where λ is a regularization parameter that controls the trade-off between fidelity to the original image and the amount of denoising.

5.3.1. Defense against FGSM Adversarial Attack

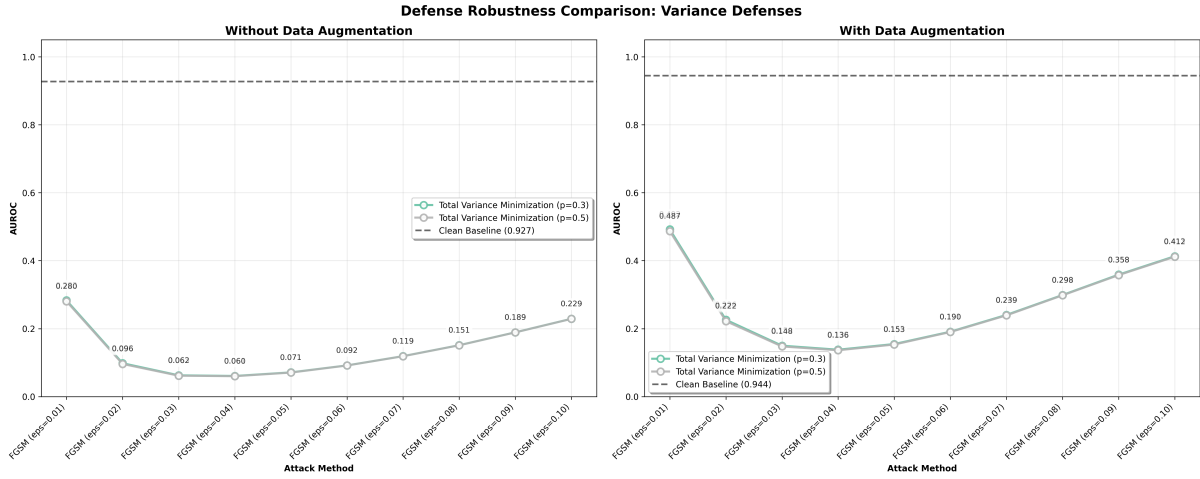


Figure 9: Classification accuracy under FGSM adversarial attacks with total variation minimization defense. Results compare performance with and without data augmentation across different p values

Figure 10 assesses the robustness provided by Total Variance Minimization (TVM) against FGSM attacks. The defense proves to be largely ineffective, causing a severe drop in AUROC even for the weakest attacks. This limited effectiveness may be attributed to the low number of iterations; `max_iter` was significantly reduced from the suggested value of 10 to 2 due to the high computational cost and the large number of experimental runs required. Both tested probability settings yield nearly identical results, with data augmentation providing only a marginal benefit. Notably, the plots exhibit a U-shaped performance curve, where accuracy begins to recover at higher epsilon values ($\text{eps} > 0.04$). We attribute this effect to the same cause of the one seen in the same circumstances in the spatial smoothing defense strategy test.

5.3.2. Defense against PGD adversarial attack

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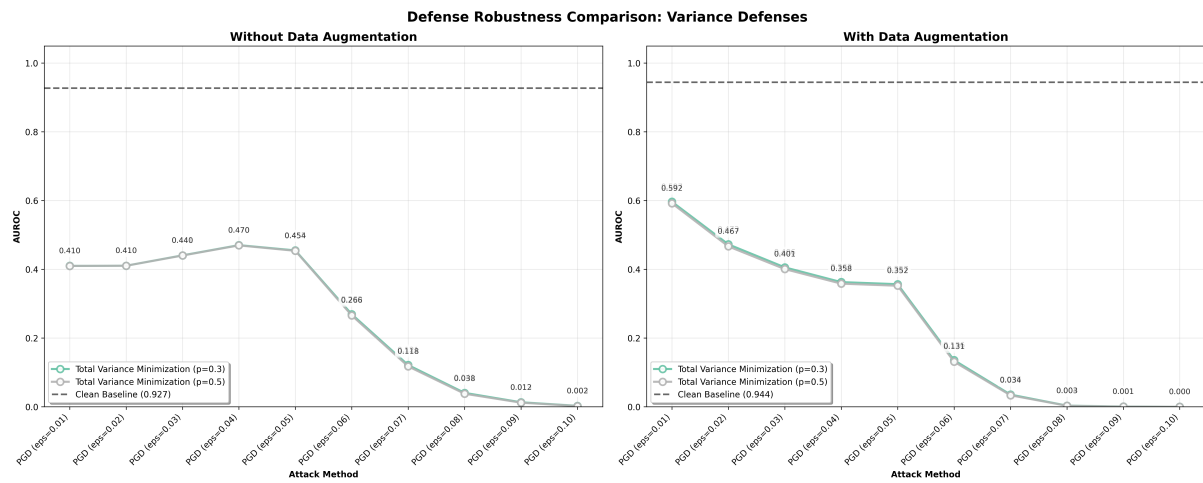


Figure 10: Classification accuracy under PGD adversarial attacks with total variance minimization defense. Results compare performance with and without data augmentation across different p values

5.3.3. Impact on the baseline performance

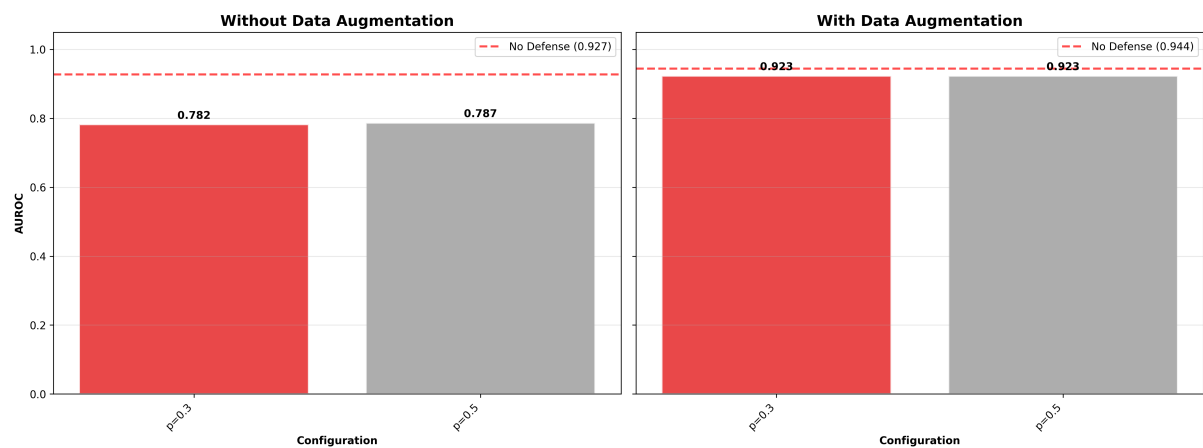


Figure 11: Impact of the total variance minimization defense on clean image classification accuracy. The plot compares performance on unprocessed clean images (red dotted line) versus images preprocessed with this defense at different p values

5.4. JPEG Compression

Simple, but surprisingly effective defense that reduces adversarial perturbations by quantizing the pixel values in the frequency domain. JPEG works by:

1. Transforming image patches using the Discrete Cosine Transform (DCT)
2. Quantizing the frequency components
3. Entropy coding for efficient storage

Second step is irreversible while decompressing the image, which is what makes JPEG useful for adversarial defense: it discards small perturbation.

Given original image x , compressed and then decompressed image defined as:

$$x_{jpeg} = JPEG^{-1}(JPEG(x))$$

Moreover, this operation can "return" image into classifier's training dataset distribution, since train images could also be JPEG files with characteristic JPEG compression artifacts.

5.4.1. Defense against FGSM Adversarial Attack

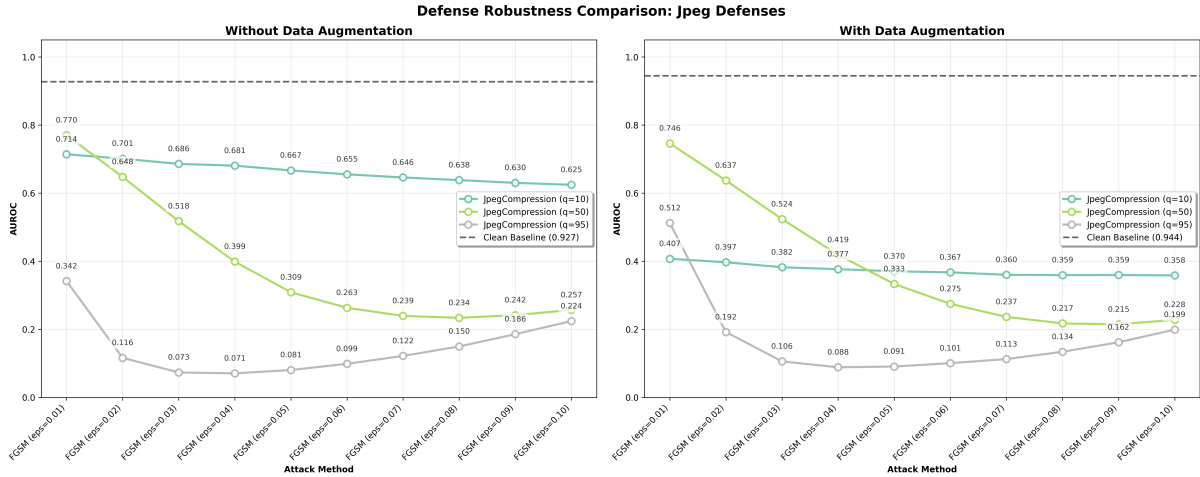


Figure 12: Classification accuracy under FGSM adversarial attacks with the Jpeg compression defense. Results compare performance with and without data augmentation across different quality (q) values

Figure 12 shows the performance of JPEG compression as a defense against FGSM attacks, evaluated at three different quality levels ($q=10$, $q=50$, and $q=95$). The results highlight a critical trade-off between compression strength and robustness. A very low quality setting ($q=10$, teal line), which corresponds to high compression, provides the most stable defense. Although it slightly degrades performance on weakly attacked images, its AUROC remains remarkably consistent as the attack strength increases. This is because aggressive compression discards significant high-frequency information, effectively filtering out the adversarial noise. Conversely, high-quality compression ($q=95$, gray line) is largely ineffective; its minimal compression fails to remove the perturbations, leading to the worst performance. A medium quality ($q=50$, yellow-green line) offers a compromise, performing well against weak attacks but degrading more sharply than the $q=10$ setting. Notably, the less effective compression settings ($q=50$ and $q=95$) exhibit a performance recovery at higher epsilon values, a phenomenon already seen in both spatial smoothing and total variation minimization defenses. As with other defenses, the model trained with data augmentation demonstrates superior overall robustness.

5.4.2. Defense against PGD adversarial attack

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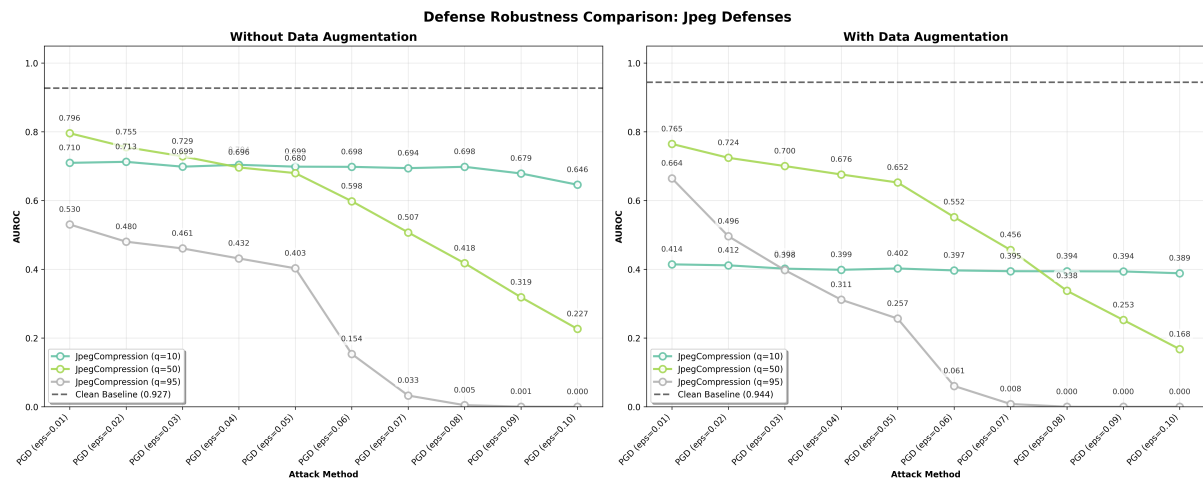


Figure 13: Classification accuracy under PGD adversarial attacks with Jpeg compression defense. Results compare performance with and without data augmentation across different quality (q) values

5.4.3. Impact on the baseline performance

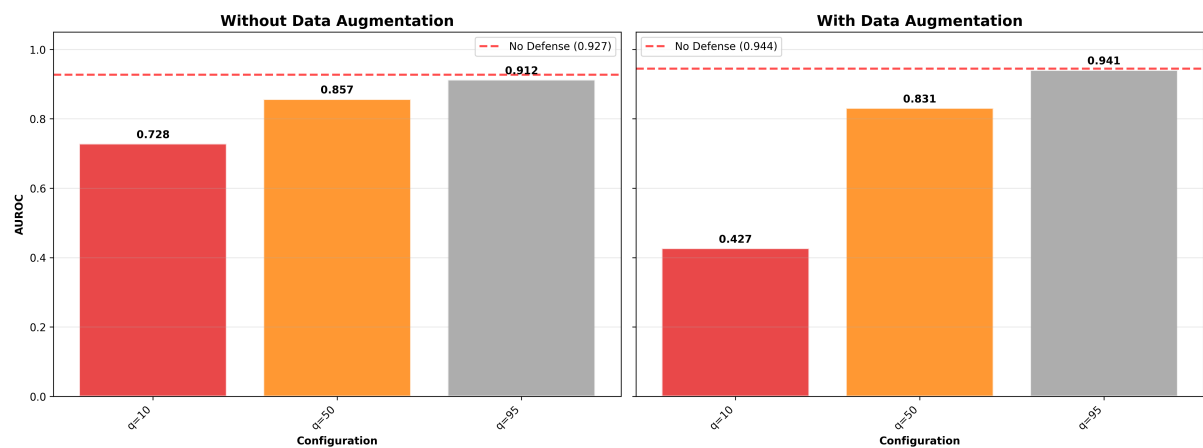


Figure 14: Impact of the Jpeg compression defense on clean image classification accuracy. The plot compares performance on unprocessed clean images (red dotted line) versus images preprocessed with this defense at different quality (q) values

5.5. Comparison between the best defense technique

6. Conclusion

7. Limitations

While this study provides a systematic evaluation of adversarial robustness within a defined scope, it is important to acknowledge its limitations. Our experimental setup was intentionally constrained to ensure a focused and reproducible analysis, and the findings should be interpreted within this context.

- **Scope of Adversarial Attacks:** Our investigation was limited to two well-established, gradient-based evasion attacks: FGSM and PGD. We did not explore other attack paradigms, such as black-box attacks (which do not require gradient information), score-based attacks, or decision-based attacks. Furthermore, our analysis does not cover other threat types like data poisoning or model extraction. The effectiveness of the evaluated defenses may differ against a broader spectrum of adversarial methods.
- **Non-Adaptive Defense Evaluation:** A significant limitation is that our defenses were evaluated under a non-adaptive gray-box scenario. In this setting, the attacker is unaware of the defense mechanism. A more sophisticated adversary could develop an *adaptive attack* by crafting perturbations specifically designed to bypass the known preprocessing step (e.g., by differentiating through the smoothing function). The resilience of simple input transformation defenses often diminishes significantly against such adaptive threats.
- **Focus on Preprocessing Defenses:** This work exclusively examines input preprocessing techniques as a defense strategy. We did not investigate other prominent defense families, such as **adversarial training**, which is widely considered one of the most effective methods for improving intrinsic model robustness. Similarly, we did not explore certified defenses, which provide mathematical guarantees of robustness within a specific threat model, or detection-based methods that aim to identify and reject adversarial inputs.
- **Model and Dataset Specificity:** The experiments were conducted using a single architecture (ResNet-18) on a single dataset (PatchCamelyon). The performance of attacks and defenses is known to be dependent on both the model architecture and the data distribution. Therefore, our findings on histopathological images may not directly generalize to other medical imaging modalities (e.g., X-rays, MRI, CT scans) or different network architectures (e.g., Vision Transformers), which may exhibit unique vulnerabilities and feature sensitivities.

We explicitly state that we operated within these settings to provide a clear and foundational benchmark on the trade-offs between standard data augmentation, attack potency, and the utility of common preprocessing defenses. Future work should aim to address these limitations by exploring adaptive attacks, incorporating adversarial training, and extending the analysis to a wider range of models and medical domains.

Declaration on Generative AI

During the preparation of this work, the author(s) used OpenAI-GPT-4 in order to: Grammar and spelling check. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

References

- [1] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken, C. I. Sánchez, A survey on deep learning in medical image analysis, *Medical Image Analysis* 42 (2017) 60–88.

- [2] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, 2015. URL: <https://arxiv.org/abs/1512.03385>. `arXiv:1512.03385`.
- [3] B. S. Veeling, J. Linmans, J. Winkens, T. Cohen, M. Welling, Rotation equivariant CNNs for digital pathology (2018). `arXiv:1806.03962`.
- [4] I. J. Goodfellow, J. Shlens, C. Szegedy, Explaining and harnessing adversarial examples, 2015. URL: <https://arxiv.org/abs/1412.6572>. `arXiv:1412.6572`.
- [5] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, A. Vladu, Towards deep learning models resistant to adversarial attacks, 2019. URL: <https://arxiv.org/abs/1706.06083>. `arXiv:1706.06083`.
- [6] G. K. Dziugaite, Z. Ghahramani, D. M. Roy, A study of the effect of jpg compression on adversarial images, 2016. URL: <https://arxiv.org/abs/1608.00853>. `arXiv:1608.00853`.
- [7] W. Xu, D. Evans, Y. Qi, Feature squeezing: Detecting adversarial examples in deep neural networks, in: Proceedings 2018 Network and Distributed System Security Symposium, NDSS 2018, Internet Society, 2018. URL: <http://dx.doi.org/10.14722/ndss.2018.23198>. doi:10.14722/ndss.2018.23198.
- [8] C. Guo, M. Rana, M. Cisse, L. van der Maaten, Countering adversarial images using input transformations, 2018. URL: <https://arxiv.org/abs/1711.00117>. `arXiv:1711.00117`.

Appendix

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